

Takumi **Shinohara**, Ph.D.

Postdoctoral Researcher

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Employment

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| May 2025 – present | Postdoctoral Researcher, KTH Royal Institute of Technology , Sweden
<i>Research on learning and control of networked controlled systems with Prof. Henrik Sandberg and Prof. Karl H. Johansson.</i> |
| Oct. 2024 – Mar. 2025 | Visiting Researcher, Keio University , Japan
<i>Research on secure control systems with Prof. Toru Namerikawa.</i> |
| Apr. 2018 – Apr. 2025 | Consultant, Mitsubishi Research Institute, Inc. , Japan
<i>Research, study, and consult on the cybersecurity policy for the Japanese Government (e.g., METI, MIC, and NISC) and consult with private companies on cybersecurity issues.</i> |

Education

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| Sep. 2021 – Sep. 2024 | Ph.D. in Engineering from Keio University, Japan
Thesis Title: <i>Secure state estimation under sensor attacks</i>
Advisor: Prof. Toru Namerikawa |
| Apr. 2016 – Mar. 2018 | M.Sc. in Engineering from Keio University, Japan
Thesis Title: <i>Zero-stealthy attacks in cyber-physical systems and secure state estimation in adversarial environments</i>
Advisor: Prof. Toru Namerikawa |

List of Publications

Preprints

- [P1] [Takumi Shinohara](#), Karl H. Johansson, and Henrik Sandberg, "Detection and identification of sensor attacks using partially attack-free data," *arXiv preprint*, arXiv:2510.02183.

Journal Papers

- [J1] [Takumi Shinohara](#), Karl H. Johansson, and Henrik Sandberg, "Distributed resilient state estimation and control with strategically implemented security measures," *IEEE Transactions on Automatic Control*, 2026. (accepted for publication as a Full Paper)
- [J2] [Takumi Shinohara](#) and Toru Namerikawa, "Relationship between the number of agents and sparse observability index," *IEEE Open Journal of Control Systems*, vol. 4, pp. 144–155, 2025.
- [J3] [Takumi Shinohara](#) and Toru Namerikawa, "Optimal security investment problem for secure state estimation on cyber-physical systems," *IEEE Transactions on Automatic Control*, vol. 70, no. 2, pp. 1244–1251, 2025.
- [J4] [Takumi Shinohara](#) and Toru Namerikawa, "Optimal resilient sensor placement problem for secure state estimation," *Automatica*, vol. 160, 111454, 2024.
- [J5] [Takumi Shinohara](#) and Toru Namerikawa, "Distributed secure state estimation with a priori sparsity information," *IET Control Theory & Applications*, vol. 16, no. 11, pp. 1086–1097, 2022.
- [J6] [Takumi Shinohara](#), Toru Namerikawa, and Zhihua Qu, "Resilient reinforcement in secure state estimation against sensor attacks with a priori information," *IEEE Transactions on Automatic Control*, vol. 64, no. 12, pp. 5024–5038, 2019.
- [J7] [Takumi Shinohara](#) and Toru Namerikawa, "Reach set-based secure state estimation against sensor attacks with interval hull approximation," *SICE Journal of Control, Measurement, and System Integration*, vol. 11, no. 5, pp. 399–408, 2018.
- [J8] [Takumi Shinohara](#) and Toru Namerikawa, "Perfect stealthy attacks in cyber-physical systems," *Transactions of the Society of Instrument and Control Engineers*, vol. 54, no. 3, pp. 309–319, 2018. (in Japanese)
- [J9] [Takumi Shinohara](#) and Toru Namerikawa, "On the vulnerabilities due to manipulative zero-stealthy attacks in cyber-physical systems," *SICE Journal of Control, Measurement, and System Integration*, vol. 10, no. 6, pp. 563–570, 2017.
- [J10] Takashi Irita, [Takumi Shinohara](#), and Toru Namerikawa, "Detection of replay attack on smart grid with code signal and bargaining game," *Transactions of the Society of Instrument and Control Engineers*, vol. 52, no. 9, pp. 498–506, 2016. (in Japanese)

Referred Conference Papers

- [C1] [Takumi Shinohara](#), Henrik Sandberg, and Karl H. Johansson, "Data-Driven Control from Poisoned Data: Fundamental Limitations and Secure DeePC," in *Proc. 65th IEEE Conference on Decision and Control*, Honolulu, HI, 2026. (accepted for presentation)
- [C2] Enno Breukelman, [Takumi Shinohara](#), Joowon Lee, and Henrik Sandberg, "Data-driven unknown input reconstruction for MIMO systems with convergence guarantees," in *Proc. 65th IEEE Conference on Decision and Control*, Honolulu, HI, 2026. (accepted for presentation)
- [C3] Kazuki Maeda, [Takumi Shinohara](#), Toru Namerikawa, "Sensor placement for distributed control systems under stealthy attacks," in *Proc. 2026 SICE Festival with Annual Conference*, Yokohama, Japan, 2026. (accepted for presentation)
- [C4] [Takumi Shinohara](#), Karl H. Johansson, and Henrik Sandberg, "Security index from input/output data: Theory and computation," in *Proc. IFAC World Congress 2026*, Busan, Korea, 2026. (accepted for presentation)
- [C5] [Takumi Shinohara](#), Karl H. Johansson, and Henrik Sandberg, "Data-driven resilience assessment against sparse sensor attacks," in *Proc. 2026 American Control Conference*, New Orleans, LA, 2026, pp. 2447–2452.
- [C6] [Takumi Shinohara](#) and Toru Namerikawa, "Secure state estimation, attack detection, and safe control against sensor attacks with side initial state information," in *Proc. 64th IEEE Conference on Decision and Control*, Rio de Janeiro, Brazil, 2025, pp. 2578–2583.
- [C7] [Takumi Shinohara](#) and Toru Namerikawa, "Security measure implementation for distributed state estimation," in *Proc. 5th IFAC Workshop on Cyber-Physical Human Systems*, Antalya, Türkiye, 2024.
- [C8] [Takumi Shinohara](#) and Toru Namerikawa, "Secure state estimation for multi-agent systems: On the relationship between the number of agents and system resilience," in *Proc. 2023 American Control Conference*, San Diego, CA, 2023, pp. 1006–1011.
- [C9] [Takumi Shinohara](#) and Toru Namerikawa, "Reach set-based attack resilient state estimation against omniscient adversaries," in *Proc. 2018 American Control Conference*, Milwaukee, WI, 2018, pp. 5813–5818.
- [C10] [Takumi Shinohara](#) and Toru Namerikawa, "Manipulative zero-stealthy attacks in cyber-physical systems: Existence space of feasible attack objectives," in *Proc. 1st IEEE Conference on Control Technology and Applications*, Kohala Coast, HI, 2017, pp. 1123–1128.
- [C11] [Takumi Shinohara](#) and Toru Namerikawa, "SLAM for a small UAV with compensation for unordinary observations and convergence analysis," in *Proc. 2016 55th Annual Conference of the Society of Instrument and Control Engineers of Japan (SICE)*, Tsukuba, Japan, 2016, pp. 1252–1257.

Honors and Awards

- Travel scholarship, Karl Engver's Foundation, 2025.
- Wallenberg Postdoctoral Scholarship, Knut and Alice Wallenberg Foundation, 2025.
- Fujiwara Award (Best Ph.D. student award), Keio University, 2025.
- Best presentation award, 65th Japan Joint Automatic Control Conference, 2022.
- Best presentation award, 59th Japan Joint Automatic Control Conference, 2016.

Review Activities

- *Journals*
 - IEEE Transactions on Automatic Control
 - IEEE Transactions on Control of Network Systems
 - IEEE Transactions on Control Systems Technology
 - IEEE Transactions on Cybernetics
 - IEEE Transactions on Intelligent Transportation Systems
 - IEEE Control Systems Letters
 - Automatica
 - Annual Reviews in Control
 - IEEE Journal on Selected Areas in Communications
 - IEEE Journal of Selected Areas in Sensors
 - ACM Transactions on Cyber-Physical Systems
- *Conferences*
 - IEEE Conference on Decision and Control
 - IEEE Conference on Control Technology and Applications
 - IFAC World Congress
 - American Control Conference
 - IEEE Vehicular Technology Conference